

ORIGINAL ARTICLE

One-day session LINAC-based stereotactic radiosurgery of posterior uveal melanoma

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PURPOSE. *LINAC-based stereotactic radiosurgery (SRS) of posterior uveal melanoma is a conservative method to treat uveal melanoma.*

METHODS. *This was a retrospective clinic-based study of patients with posterior uveal melanoma in stage T2/T3 who underwent 1-day session SRS at LINAC accelerator or SRS plus combined methods from 2001 to 2008.*

RESULTS. *Thirty-nine patients with posterior uveal melanoma were treated with SRS (age 25-80 years, median 54 years). Median tumor volume at baseline was 0.6 cm³ (range 0.2-1.3 cm³). The therapeutic dose (TD) was 35.0 Gy, median of maximal dose applied was 49.0 Gy (range 37.0-60.0 Gy). Patient data were analyzed in groups: group 1, single SRS irradiation; group 2, SRS with subsequent endoresection or cyclectomy or additional transpupillary thermotherapy (TTT) or brachytherapy by Ru106 plaques; group 3a, enucleation after single SRS; group 3b, enucleation after SRS and endoresection/cyclectomy or TTT or brachytherapy Ru106. In patients with visual acuity of 20/40 or better, the median rate of best-corrected visual acuity (BCVA) decline was higher than that of the total and significantly higher than the rate of decline in the complementary group of patients with BCVA less than 20/40 ($p=0.0077$; Mann-Whitney U test).*

CONCLUSIONS. *One-step LINAC-based SRS with a single dose 35.0 Gy is a method to treat middle-stage posterior uveal melanoma and to preserve the eye globe or as the first step of combined methods: irradiation before endoresection or cyclectomy.*

KEY WORDS. *LINAC-based stereotactic radiosurgery, Uveal melanoma*

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INTRODUCTION

Uveal melanoma is the most common (over 95% of the total) and most aggressive type of intraocular tumor in adults; up to 50% of patients die within 15 years after enucleation or other therapeutic methods (1).

Age and volume (size) of the tumor are prognostic indicators following enucleation for posterior uveal melanoma (2). The incidence of intraocular tumors varies from 0.2 to 1.0/100,000 inhabitants (3).

The recorded data from Slovak regions correspond to the values reported from other countries and regions of Europe. In less populated countries like Slovakia, where the whole population is slightly above 5.5 million inhabitants, the number of new cases of uveal melanoma diagnosed per year varies from 10 to 25. As issued by the Slovak National Oncology Register, the incidence in Slovakia is 0.2 to 0.6/100,000 inhabitants (4).

Over the past 3 decades, diagnostic methods have improved and radiotherapy (external beam, charged particle,

or brachytherapy) has replaced enucleation as the preferred treatment for most patients with uveal melanoma. The desire to improve survival and preserve vision in patients with uveal melanoma has stimulated the development of alternative therapies. Different radiation modalities are currently in use in treatment of posterior uveal melanoma. One of the methods of conservative approach is stereotactic radiosurgery (SRS).

Single irradiation of the tumor itself is a new approach that has been shown to achieve ultrasonic tumor regression in a similar fashion to brachytherapy. Stereotactic radiosurgery of extracerebral lesions like uveal melanoma has been invented in the last 2 decades and is an alternative treatment for middle and large posterior choroidal melanoma. With radiotherapy, eye salvage is achieved, and, particularly for cases in which the tumor is located away from the optic disc or macula, useful vision can be retained after treatment (3).

Some authors prefer irradiation before enucleation for large uveal melanoma (5-7). They used SRS with a single fraction administered with a precise spatial accuracy using a collimating system. Image fusion of contrast-enhanced magnetic resonance imaging (MRI) and computed tomography (CT) is used for treatment planning coordinates. Stereotactic radiosurgery in treatment of uveal melanoma in Slovakia is performed on linear accelerator LINAC. In a retrospective study in Slovakia no survival difference attributable to stereotactic irradiation or enucleation of uveal melanoma has been demonstrated. Enucleation after SRS in 7 patients occurred at an interval of 6 to 24 months after SRS. A small difference is possible, but a clinically meaningful difference in mortality rates, whether from all causes or from metastatic melanoma, is unlikely (8).

High rates of local control can be achieved, with 5-year control rates exceeding 95% in patients treated with charged particles (9). Proton beam radiotherapy with a 62 MeV cyclotron achieves high rates of local tumor control and ocular conservation, with visual outcome depending on tumor size and location (10).

Large, prospective, randomized trials were designed to compare mortality figures for medium-sized melanomas treated by brachytherapy or enucleation (1, 11). The results showed no difference in mortality rates between the 2 treatment groups after a maximum of 12 years of follow-up (7). LINAC based radiosurgery is a novel treatment approach for choroidal melanoma. In this study, we evaluated local failure

leading to enucleation as an endpoint in patients treated by SRS with long-term follow-up having accrued at the time of analysis, and assessed the treatment of posterior uveal melanoma by 1-day session of LINAC SRS.

The first aim of our study was to assess after-treatment BCVA decline in patients with posterior uveal melanoma treated with SRS in a 6-month interval 24 months after SRS.

The second aim was to find out whether the group of patients with better initial visual acuity at the beginning of treatment also would have a better chance of preserved vision. The observed after-treatment decline in BCVA was a 24-month interval after the treatment.

The third aim was monitoring of the tumor regression by measuring the maximum elevation by B-scan ultrasound in the group of patients with single irradiation (SRS) in an interval 12 and 24 months after the therapy.

METHODS

A retrospective analysis was undertaken for patients with uveal melanoma treated in 2001-2008 by SRS. We reviewed the records of 84 patients with ciliary body or choroidal melanoma treated in this period. From the group of 84 patients, 44 patients underwent primary enucleation (52.4%) and 40 patients underwent SRS as a first-step treatment (47.6%). The diagnosis was established on the basis of ophthalmologic examination, ultrasound, CT, or MRI examination. Metastatic intraocular tumors were not included in the analyzed cohort; juxtapapillary tumors and melanocytomas also were excluded.

Patients were not randomized either to radical or to "conservative" procedure, but the treatment was determined exclusively on a case-by-case basis. Tumor volume, maximum elevation, localization presence of secondary retinal detachment, general status, age, gender, the functional tests (visual acuity, perimeter), and the visual acuity of the other eye all were taken into consideration. The patient was involved in the selection of the therapeutic procedure after possible post-operative complications had been explained to him. Immobilization of the treated eye was achieved by mechanical fixation to the stereotactic Leibinger frame. For fixation of the eye globe we used silk 6/0 sutures. Sutures were placed through conjunctiva and through the tendons of 4 direct extraocular muscles (Fig. 1, A and B). This step of surgery was done at the Department of Ophthalmol-

ogy the day before SRS using local anesthesia. The next morning the stereotactic frame was fixed to the head and the sutures were tied to the stereotactic frame at the St. Elisabeth Department of Stereotactic Radiosurgery. In our study, we used model Clinac 600 C/D of the manufacturer Varian (system Aria, planning system Corvus version 6.2 verification IMRT OmniPro) with 6 MeV X by rigid immobilization of the eye to the Leibinger frame. The patient underwent CT and MRI examination with the fixed eye to the frame. The stereotactic treatment planning after fusion of CT and MRI was optimized according to the critical structures (lens, optic nerve, and also lens and optic nerve at the contralateral side, chiasm). The best plan was applied for therapy at C LINAC accelerator (Fig. 2, A and B). In the afternoon, the patient underwent irradiation and sutures and frame were removed. The next morning the patient underwent slit-lamp examination, ophthalmoscopy, intraocular pressure (IOP) measurement and was released for home treatment with local therapy (eyedrops: antibiotics, lubricant).

The planned therapeutic dose in SRS was 35.0 Gy, TDmin dose to the margin of the lesion varied from 35.0 to 38.0 Gy, TDmax 37.0-60.0 Gy. The therapeutic dose was 35.0 Gy in 37 patients, in 2 cases 38.0 Gy. We used planning treatment volume 95% isodose planning. The doses to the critical structures were below 8.0 Gy for the optic nerve and the optic disc and 10.0 Gy to the anterior segment of the eye. Patients with melanocytoma or patients with juxtapapillary melanomas were excluded from the study. In 8 patients with tumor location 2 PD from the optic disc the dose to disc and surrounding structures was 12.0 Gy.

The record for each patient included the age at treatment, tumor size, tumor volume, the maximum height of the tumor by A, B-scan ultrasound, the presence and the extent of secondary retinal detachment, and signs of extrascleral extension. Tumor volume was calculated in each SRS group patient directly by computer after CT and MRI examination as the step of SRS procedure and was involved to the stereotactic planning scheme. Tumor volume calculation was based on the region of interest of the tumor and 3-D reconstruction was done.

Tumors were divided into 3 groups, as follows: small, 4 resp. 5 mm of maximal elevation; middle, 4-8 mm; and large, over 8 mm.

In the group of single stereotactic irradiation (group 1), the elevation of the tumor was observed in 6-month intervals by B-scan ultrasound by one ophthalmologist. We compared

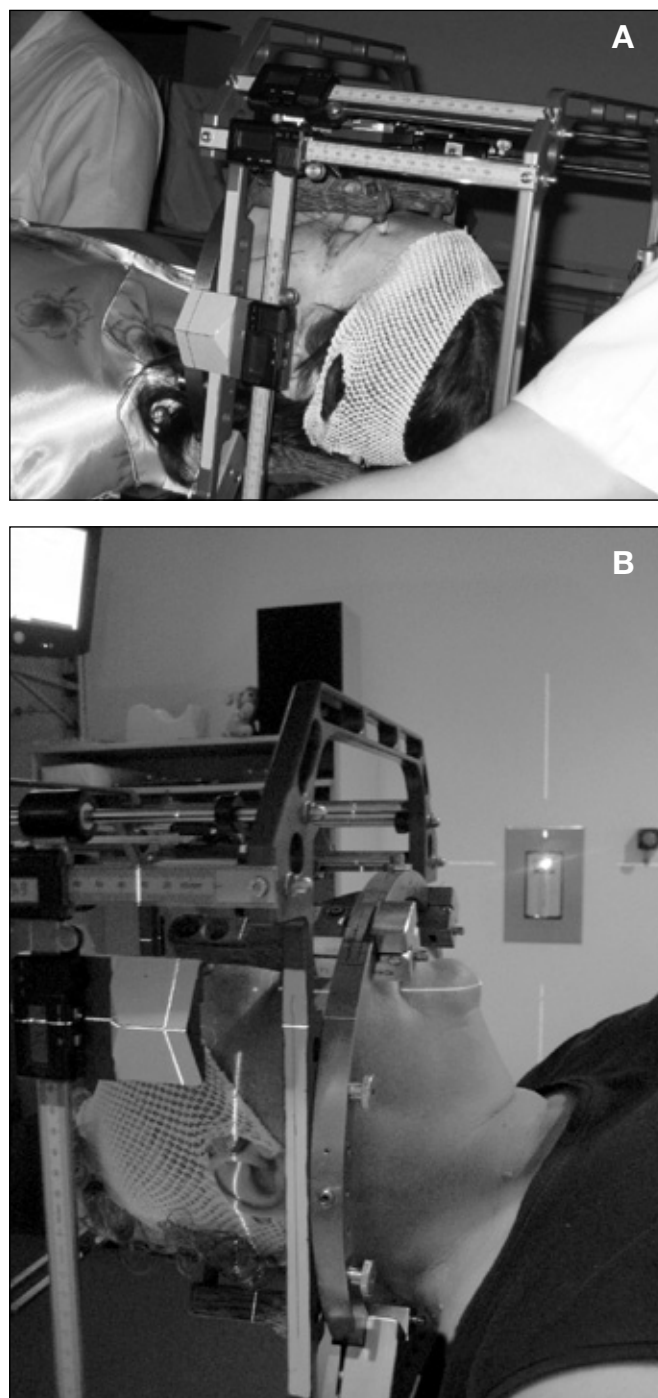


Fig. 1 - (A, B) Patients positioning before treatment.

tumor regression by measuring the maximum elevation by B-scan ultrasound in the group of 25 patients with single irradiation (SRS) 12 and 24 months after the therapy.

Patients were recommended to their oncologist regularly in 6-month intervals for liver ultrasound, abdominal ultra-

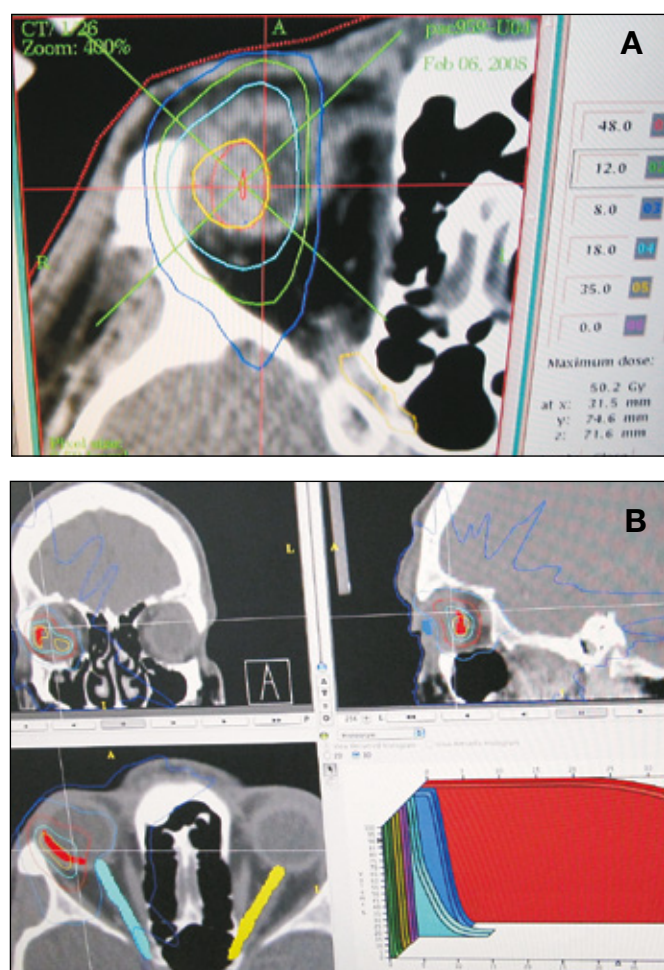


Fig. 2 - (A, B) Stereotactic planning schemes.

sound, and liver function test, and had chest X-ray once per year to confirm or exclude the presence of metastases. In individual cases, they were recommended for brain CT or positron emission tomography.

RESULTS

Overall, 84 patients with posterior uveal melanoma were treated from 2001-2008 at the Department of Ophthalmology, Comenius University, Bratislava.

Primary enucleation in the cohort was indicated in 44 (52.4%) patients and 1-day session SRS by LINAC C in 40 (47.6%) patients.

One patient treated by 1-day session SRS in that period dropped out of the data processing because the patient

was no longer taking part in further observation. For the further analysis we used the rest: the data of 39 patients.

In this observed group of 39 patients who underwent SRS, there were 22 male (56.4%) and 17 female (43.6%) patients. The data of the patients were analyzed in groups: group 1, stereotactic irradiation–single irradiation (25 patients); group 2, stereotactic irradiation with subsequent endoresection/cyclectomy or transpupillary thermotherapy (TTT) or brachytherapy Ru106 (14 patients).

The additional analysis related to enucleation was done for group 3a (enucleation after SRS–single irradiation) (5 patients) and group 3b (enucleation after SRS and endoresection/cyclectomy or TTT or brachytherapy Ru106) (7 patients).

The group 1 and 2 overview is summarized in Table I, including the total median age, the median age of women and men, and other descriptive characteristics (tumor volume, total dose TDmax related to the follow-up in months). The values are reported as median (lower quartile–upper quartile). In group 1, patients with single irradiation of the tumor, median age was 53 years; in group 2, patients treated with combined methods (SRS plus endoresection resp. cyclectomy), median age was higher, at 56 years.

Visual acuity outcome and results

The decline of vision according to the pretreatment best-corrected visual acuity (BCVA) by Snellen chart at 6, 12, and 24 months in 39 patients is shown in Table II.

The data showed that nearly a quarter of patients had initial BCVA of 20/40 or better; BCVA decline after 6 months showed that in nearly 49% it had worsened to less than 20/40. Best-corrected visual acuity significantly declined 6-12 months after the therapy. The observed after-treatment decline in BCVA was not positively associated with higher prevalence of 20/40 or better BCVA. We did not correlate the distance from the macula.

The patients in group 1 (25 patients) showed lower median rate of BCVA decline (change) by Snellen chart per 6 months if compared with group 2 (14 patients). Mann-Whitney *U* statistic did not confirm this difference in the rate of decline as significant ($p=0.44$).

Analysis of individual patient differences in the groups showed a lower number of patients with unchanged or improved visual acuity in the SRS groups (Figs. 3 and 4) in 12 months with a probability value of 0.25 (Fisher exact test). This indicates a trend towards lower decline (negative progression of the rate) in the group of patients

TABLE I - GROUP OF 39 PATIENTS WITH INTERMEDIATE OR LARGE UVEAL MELANOMA TREATED BY SRS (2001-2008)

	Total	Group 1	Group 2
No. of patients	39	25	14
Female	17	8	9
Male	22	17	5
Age, y	54 (49-65)	53 (50-65)	56 (42-62)
Female	53 (49-66)	66 (53-66)	45.5 (39-59)
Male	58 (51-62)	52.5 (42-62)	62 (60-72)
Volume, cm ³	0.6 (0.4-0.8)	0.6 (0.4-0.7)	0.7 (0.5-0.8)
Female	0.6 (0.4-0.8)	0.4 (0.3-0.4)	0.7 (0.45-0.8)
Male	0.65 (0.4-0.9)	0.4 (0.3-0.6)	0.7 (0.5-0.8)
Total dose TDmax, Gy	49 (45-50)	46 (41-49)	50 (48-51)
Female	50 (48-51)	48 (45-50)	50 (48.5-51.5)
Male	47.5 (41-59.1)	45.5 (40-48)	50 (48-50)
Follow-up, mo	36	30	54

SRS = stereotactic radiosurgery.

TABLE II - BEST-CORRECTED VISUAL ACUITY (BCVA) AT 6, 12, AND 24 MONTHS

BCVA values	No. of patients (%)	BCVA (feet)	Snellen chart
Before therapy	9 (23)	≥20/40	≥0.5
	19 (49)	<20/40 and ≥20/200	<0.5 and ≥0.1
	11 (28)	<20/200	<0.1
6 months after SRS	7 (18)	≥20/40	≥0.5
	19 (49)	<20/40 and ≥20/200	<0.5 and ≥0.1
	13 (33)	<20/200	<0.1
12 months after SRS	2 (5)	≥20/40	≥0.5
	12 (31)	<20/40 and ≥20/200	<0.5 and ≥0.1
	25 (64)	<20/200	<0.1
24 months after SRS	1 (3)	≥20/40	≥0.5
	11 (28)	<20/40 and ≥20/200	<0.5 and ≥0.1
	27 (69)	<20/200	<0.1

SRS = stereotactic radiosurgery.

treated only with SRS. This should be found by analyzing higher numbers of patients.

Multiple linear regression did not reveal any association between the rate of decline and age ($p=0.84$) or tumor volume ($p=0.33$) in these groups.

Therefore, based on the regression analysis, the estimated median decline in the groups can be considered as independent of age and tumor volume baseline characteristics. The prognosis of patients who did not need an additional intervention (group 1) was significantly better than that in

the patients with subsequent endoresection or cyclectomy (group 2). This result was not related to age or tumor volume recorded at initial examination.

Tumor regression in group 1 by B-scan ultrasound

Patients with single irradiation by SRS in group 1 were divided into 3 subgroups according to the maximum elevation before irradiation: small, 4 to 5 mm = 4 cases (16%); middle, up to 8 mm high = 15 cases (60%); large, over 8 mm

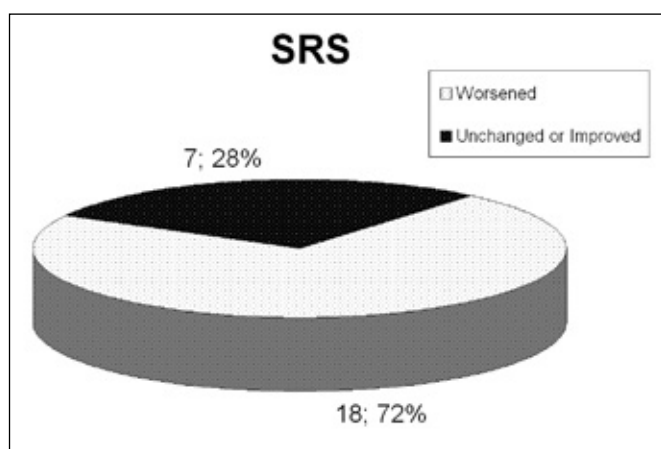


Fig. 3 - Ratio of individual patient differences in the stereotactic radiosurgery (SRS) group 12 months after treatment.

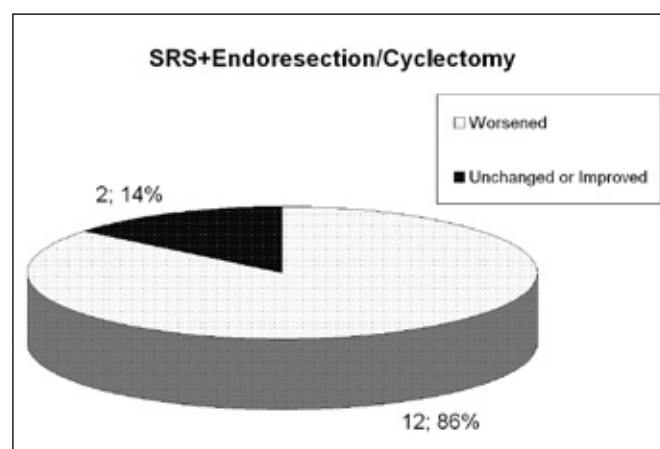


Fig. 4 - Ratio of individual patient differences in the stereotactic radiosurgery (SRS) + endoresection/cyclectomy group 12 months after treatment.

= 6 cases (24%). Tumor regression after the treatment in 6-month intervals 12 months after the therapy showed that in the group of small tumors in all of the patients there was no presence of increase of the elevation, but in 24 months there was sign of tumor regression in 3 cases (75%). In the middle stage group 12 months after the therapy there was no sign of tumor elevation regression, but in the 24 months after the therapy tumor regression more than 1 mm of the maximum pretreatment elevation was found in 4 cases (27%). In the third subgroup of large tumors, there was no sign of tumor regression according to ultrasound results in the 12-month or in the 24-month interval after the therapy.

Next we analyzed patients who underwent enucleation (group 3, a and b).

Group 3a, secondary enucleation after SRS as single treatment, showed the following: in patients with SRS as a single method of treatment, enucleation was necessary in 5 patients (20%) in 3 to 5 years after irradiation; in all of the cases the tumors had pretreatment maximum elevation 10 mm and more; the tumor volume was 0.5 to 0.7 mm³ (average 0.9 mm³). In 3 cases, the tumor was arising from the ciliary body. Histopathologic findings showed spindle cell melanoma in 3 patients and mixed cell melanoma in 2 patients. There was no presence of optic nerve infiltration in the enucleated eye globes.

Group 3b, secondary enucleation after SRS plus endoresection or cyclectomy and TTT or brachytherapy Ru106 after endoresection or cyclectomy results, showed the following: in patients with SRS plus endoresection or

cyclectomy and TTT or brachytherapy after endoresection or cyclectomy, secondary enucleation was necessary in 7 cases (50% of all the patients in group 2) in 12-24 months after the primary therapy due to the complications: secondary neovascular glaucoma (see Fig. 5) or tumor relapse. The tumors had pretreatment maximum elevation 6-10 mm and the tumor volume was 0.6-1.0 mm³ (average 0.9 mm³). In this group of patients with secondary enucleation after SRS, 3 of the patients underwent pars plana vitrectomy with incomplete endoresection of the tumor plus silicone oil, but the enucleation was necessary due to complications such as relapse of the tumor, bleeding into the anterior chamber, or secondary retinal detachment.

Histopathologic findings

In the group of enucleated eye globes, in 3 cases the malignant melanoma of mixed cell type was present, in one case an epithelioid type, and in 3 cases the spindle-cell type A was confirmed. There was no presence of optic nerve infiltration in these enucleated eye globes.

DISCUSSION

In the last 3 decades, the management of patients with uveal melanoma has shifted toward globe-sparing techniques. Alternatives to radical enucleation vary from

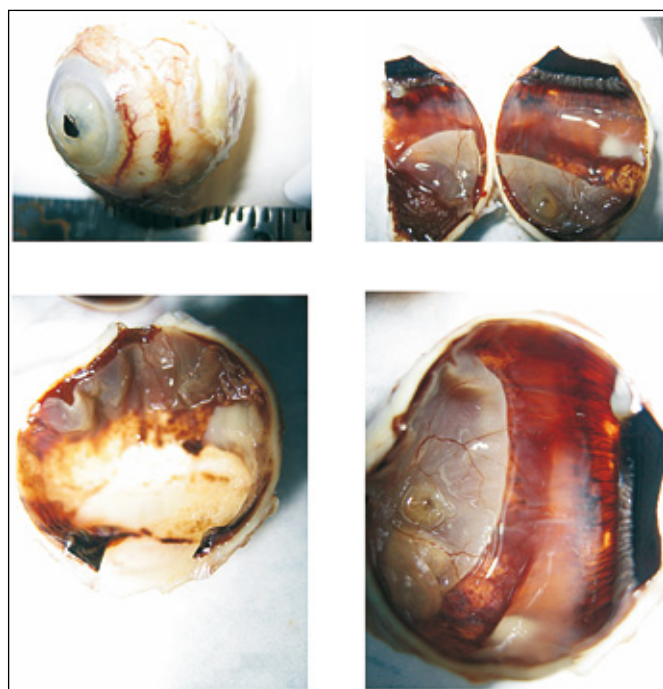


Fig. 5 - Enucleated eye globe after stereotactic radiosurgery and complete endoresection due to secondary glaucoma and keratopathy 2 years after endoresection.

observation to TTT, block-excision, endoresection with pars plana vitrectomy, brachytherapy using a variety of radioisotopes, external beam radiotherapy, charged particles, and SRS, or the methods can be combined. Stereotactic radiosurgery has recently been proposed as an alternative treatment for posterior uveal melanoma.

The therapy for each patient should be chosen in accordance with the general status of the patient and with the local findings, stage, and character of the tumor (3). The Collaborative Ocular Melanoma Study, a multicenter national trial, is intended to provide long-term data on the natural history as well as therapeutic intervention. This large, prospective, randomized trial was designed to compare mortality figures for medium-sized melanomas treated by brachytherapy or enucleation (5, 6). The results were not able to show the difference in mortality rates between the 2 treatment groups after a maximum of 12 years of follow-up (7). The study was set up in 1985, before SRS was introduced in the treatment of uveal melanoma.

Stereotactic radiosurgery is a relatively new method, so there is a need for a multicenter trial to compare the out-

comes following SRS with other methods. However, until now, no study has been performed in this topic. Studies comparing survival rates following enucleation versus newer treatment modalities, including SRS, suggested similar rates for comparable lesions and because reported local tumor control rates following SRS appear comparable, we offer SRS to patients who would otherwise require enucleation (1, 7, 11, 12).

Local control of over 95% has been reported in some studies: in the study of Dieckmann et al (13), local control was 98% after a median observation time of 33 months follow-up. The observation time is still too short to allow definitive conclusions, but their results are comparable with the 82%-98% local control rate reported by other groups after a median observation time of up to 15 years.

Seddon et al (14) described visual loss after proton beam irradiation of 33% to 47% after 1 and 2 years, respectively, for tumors located near the optic disc and fovea.

Meyer et al (15) reported in a retrospective study that irradiation of 30.0 Gy of more than 2 mm of the optic nerve head initiated an optic neuropathy.

In the study of Dieckmann et al (16), due to unfavorable tumor size and location in the vicinity of critical structures, e.g., optic nerve and macula, visual reduction was noticed in a high number of the patients. After an observation time of more than 6 months, visual acuity could be evaluated in 79 patients. In the group of 77 patients, 85.5% presented with visual acuity of 0.1 or better prior to radiotherapy. LINAC-based stereotactic irradiation for uveal melanoma is feasible and well-tolerated and can be offered to patients with medium-sized and unfavorably located uveal melanoma who are searching for an eye-preserving treatment (16).

The tumor localization is important to achieve good visual acuity result. Ruthenium 106 brachytherapy of posterior choroidal melanoma achieves good conservation of vision if the tumor does not extend close to the optic nerve or fovea (17).

Stereotactic photon therapy of uveal melanoma, based on CT and MRI, is a safe and precise treatment option. Local control was found to be excellent. Because of selection criteria, the number of patients in the study with reduced visual acuity will probably increase in the future (13).

It is important to realize that the power of a test to compare survival in 2 or more groups is related not to the total sample size but to the number of events of interest (such as deaths in this case). In other words, the survival

tests perform better when the censoring is not too heavy, and, in particular, when the pattern of censoring is similar across the different groups. A high number of right-censored data (from those patients who still were alive at the end of observation, or dropped out of the study for various reasons other than death prior to its termination) could affect the reliability of the results. Thus, the heavy censoring might complicate the estimation of the survival model, because it decreases the equivalent number of subjects exposed (at risk) at later times, reducing the effective sample sizes. Moreover, small sample sizes may further increase the effect of the assumption violation. It is not reasonable, however, to drop the selected explanatory variables from the model, since there are real-world reasons why these particular variables should remain in the final model (6, 18-20).

There have been no multicenter trials to assess dosimetry, safety, and efficacy of SRS, or to evaluate outcomes of gamma knife radiosurgery for melanoma, but data from several reported case series suggest that SRS can have similar local tumor control rate, metastasis rate, mortality rate, and complications rate when compared to brachytherapy (21-25). Recent studies have suggested that gamma knife radiosurgery and SRS may be an appropriate alternative for treating uveal melanoma in those patients in whom lesions are ineligible for conventional brachytherapy (2, 26-28). The findings in the series suggest a role of SRS in the treatment of selected cases of uveal melanoma.

Complications after certain procedures can lead to secondary neovascular glaucoma and may result in enucleation; eye retention is one of the main goals of conservative treatment (29-32).

A multivariate data analysis by employing the supervised learning techniques, in particular the algorithm known as regularized least squares, was used in study of Mosci et al (33). Their study was the largest one in Italy and they demonstrated the excellent local tumor control, survival, and eye retention rate after the proton beam irradiation therapy. According to their results, future refinements in treatment planning, dosing, and delivery could be necessary to determine visual results and complications after proton beam therapy in ocular melanoma. In the procedure of SRS, there is necessity of refinements in the planning system as well. The main issues with single-session radiotherapy are the effects of distribution and hypofractionation of the dose. Tumor size and location, e.g., closer than 2 mm to the optic disc, are the most important factors to assess clinical

evaluation of visual acuity outcome.

Identification of risk factors may reduce the rates of recurrence and lead to fewer complications, preservation of the eye, improved visual function, and, potentially, better survival outcome. The recurrence of optic neuropathy after SRS is a problem not only by intraocular tumors but also, e.g., by perichiasmal tumors stereotactic irradiation. Although rare, optic neuropathy may follow radiosurgery to lesions near the visual pathways. Careful dose planning guided by MRI with restriction of the maximal dose to the visual pathways to less than 8.0 Gy will likely reduce the incidence of this complication (34).

The same problems with visual acuity loss as in SRS are found in patients after other radiotherapy procedures, e.g., brachytherapy. In the consecutive series of patients after Ru106 brachytherapy, patients retained some useful vision in the first postoperative years and a few even got better visual acuity; however, the long-term visual outcome is poor with a continuing visual acuity loss over time. A large number of patients became blind or lost reading ability after 5 years, either because of radiation complications or secondary enucleation (35).

According to our results, tumor regression after the single SRS treatment at a 6-month interval in the first year after the therapy showed that in the group of small tumors in all of the cases there was no increase of the elevation; in 24 months, there was tumor regression in 3 cases (75%). In the middle stage group 1 year after the therapy, there was no sign of tumor elevation regression, but in a 2-year interval after the therapy tumor regression was found in 4 cases (27%); in the subgroup of large tumors, there was no sign of tumor regression in the 12- or 24-month interval after the therapy. Probably in more than 5 years, these numbers would be changed.

Stereotactic radiosurgery and fractionated stereotactic radiotherapy have emerged as promising, noninvasive treatments for uveal melanoma (36). Although, historically, melanoma has been considered a relatively radioresistant tumor, newer data have challenged this viewpoint, and radiation therapy is now considered to be a useful component of the therapeutic armamentarium for malignant melanoma. According to our results, a single 1-day session SRS with 35.0 Gy is sufficient to treat small and middle stage melanoma. According to our clinical experience, SRS as a single 1-day session with 35.0 Gy can be successfully used as irradiation before other surgical procedures (endoresection, cyclectomy).

CONCLUSIONS

Stereotactic radiosurgery is a noninvasive alternative to enucleation in the treatment of uveal melanoma with high tumor control (8). LINAC-based SRS with a single dose 35.0 Gy in conjunction with a mechanical immobilization system with 4 sutures according to our study is a highly effective method to treat middle sized uveal melanoma and to preserve the eye globe with a sufficient visual acuity. Secondary enucleation in a group of 39 patients with SRS was necessary in 7 cases 1-3 years after SRS due to secondary glaucoma and relapse of the tumor. In this subgroup of uveal melanoma, the treatment was in first step the SRS procedure and continued with endoresection or incomplete cyclectomy. In case of residual tumour mass additional TTT or episcleral brachytherapy with Ru106 plaque was performed. The observed after-treatment decline in BCVA was not positively associated with higher prevalence of better BCVA before SRS, but the anatomic result after the treatment was a preserved eye globe.

Our results justify further studies to evaluate a 1-day session procedure and its efficacy as an alternative to other irradiation therapeutic approaches. In our experience SRS with prescribed dose 35.0 Gy is viable method for patients with small and intermediate tumours with elevation up to 6 mm and volume up to 0.4 cm³. For patients with larger tumour volumes over 0.8 cm³ it is necessary to combine

SRS with another treatment modalities. This therapy is sufficient for small and intermediate tumors with elevation not over 6 mm, resp. volume up to 0.4 cm³, according to the individual stereotactic planning scheme of each patient as a single therapy procedure. One-day session SRS with 35.0 Gy is sufficient to treat small and middle stage melanoma. According to our clinical experience, SRS as a single 1-day session with 35.0 Gy can be successfully used as irradiation before other surgical procedures (endoresection, cyclectomy).

There is a need for multicenter trials to compare the outcomes following SRS in treatment of uveal melanoma.

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REFERENCES

1. Singh AD, Shields CL, Shields JA. Prognostic factors in uveal melanoma. *Melanoma Res* 2001; 11: 255-63.
2. Seregard S, Kock E. Prognostic indicators following enucleation for posterior uveal melanoma: a multivariate analysis of long-term survival with minimized loss to follow-up. *Acta Ophthalmol Scand* 1995; 73: 340-4.
3. Shields CL, Shields JA, Cater J, et al. Plaque radiotherapy for uveal melanoma: long-term visual outcome in 1106 consecutive patients. *Arch Ophthalmol* 2000; 118: 1219-28.
4. Ondrusova M, Ondrus D, Dusek L, Muzik J. National portal of cancer epidemiology in the Slovak Republic. *Bratisl Lek Listy* 2008; 109: 329-30.
5. Melia BM, Abramson DH, Albert DM, et al. Collaborative Ocular Melanoma Study Group. Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma: I: visual acuity after 3 years: COMS report no. 16. *Ophthalmology* 2001; 108: 348-66.
6. Collaborative Ocular Melanoma Study Group. Assessment of metastatic disease status at death in 435 patients with large choroidal melanoma in the Collaborative Ocular Melanoma Study (COMS): COMS report no. 15. *Arch Ophthalmol* 2001; 119: 670-6.
7. Diener-West M, Earle JD, Fine SL, et al. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: initial mortality findings: COMS Report No. 18. *Arch Ophthalmol* 2001; 119: 969-82.
8. Furdova A, Slezak P, Chorvath M, et al. No differences in outcome between radical surgical treatment (enucleation) and ste-

- reotactic radiosurgery in patients with posterior uveal melanoma. *Neoplasma* 2010; 57: 377-81.
9. Zehetmayer M, Kitz K, Menapace R, et al. Local tumor control and morbidity after one to three fractions of stereotactic external beam irradiation for uveal melanoma. *Radiother Oncol* 2000; 55: 134-44.
10. Damato B, Kacperek A, Chopra M, Campbell IR, Errington RD. Proton beam radiotherapy of choroidal melanoma: the Liverpool-Clatterbridge experience. *Int J Radiat Oncol Biol Phys* 2005; 62: 1405-11.
11. Cohen VML, Carter MJ, Kemeny A, Radatz M, Rennie IG. Metastasis-free survival following treatment for uveal melanoma with either stereotactic radiosurgery or enucleation. *Acta Ophthalmol Scand* 2003; 81: 383-8.
12. Gragoudas ES, Lane AM, Munzenrider J, Egan KM, Li W. Long-term risk of local failure after proton therapy for choroidal/ciliary body melanoma. *Trans Am Ophthalmol Soc* 2002; 100: 43-8; discussion 48-9.
13. Dieckmann K, Dietmar G, Bogner J, et al. Optimizing LINAC-based stereotactic radiotherapy of uveal melanomas: 7 years clinical experience. *Int J Radiat Oncol Biol Phys* 2006; 66(Suppl): S47-52.
14. Seddon JM, Gragoudas ES, Egan KM, et al. Uveal melanomas near the optic disc or fovea: visual results after proton beam irradiation. *Ophthalmology* 1987; 94: 354-61.
15. Meyer A, Lévy C, Blondel J, et al. [Optic neuropathy after proton-beam therapy for malignant choroidal melanoma.] *J Fr Ophtalmol* 2000; 23: 543-53.
16. Dieckmann K, Georg D, Zehetmayer M, Bogner J, Georgopoulos M, Pötter R. LINAC based stereotactic radiotherapy of uveal melanoma: 4 years clinical experience. *Radiother Oncol* 2003; 67: 199-206.
17. Damato B, Patel I, Campbell B, Mayles MH, Errington RD. Visual acuity after Ruthenium 106 brachytherapy of choroidal melanomas. *Int J Radiat Oncol* 2005; 63: 392-400.
18. Augsburger JJ, Goel SD. Visual function following enucleation or episcleral plaque radiotherapy for posterior uveal melanoma. *Arch Ophthalmol* 1994; 112: 786-9.
19. Augsburger JJ, Schneider S, Freire J, Brady LW. Survival following enucleation versus plaque radiotherapy in statistically matched subgroups of patients with choroidal melanomas: results in patients treated between 1980 and 1987. *Graefes Arch Clin Exp Ophthalmol* 1999; 237: 558-67.
20. Li W, Gragoudas ES, Egan KM. Metastatic melanoma death rates by anatomic site after proton beam irradiation for uveal melanoma. *Arch Ophthalmol* 2000; 118: 1066-70.
21. De Potter P, Shields CL, Shields JA, Cater JR, Tardio DJ. Impact of enucleation versus plaque radiotherapy in the management of juxtapapillary choroidal melanoma on patient survival. *Br J Ophthalmol* 1994; 78: 109-14.
22. Langmann G, Pendl G, Klaus-Müllner G, Papaefthymiou G, Guss H. Gamma knife radiosurgery for uveal melanomas: an 8-year experience. *J Neurosurg* 2000; 93(Suppl 3): 184-8.
23. Marchini G, Gerosa M, Piovan E, et al. Gamma Knife stereotactic radiosurgery for uveal melanoma: clinical results after 2 years. *Stereotact Funct Neurosurg* 1996; 66(Suppl 1): 208-13.
24. Mueller AJ, Talies S, Schaller UC, Horstmann G, Wowra B, Kampik A. Stereotactic radiosurgery of large uveal melanomas with the gamma-knife. *Ophthalmology* 2000; 107: 1381-7; discussion 1387-8.
25. Rennie IG, Forster D, Kemeny A, Walton L, Kunkler I. The use of single fraction Leksell stereotactic radiosurgery in the treatment of uveal melanoma. *Acta Ophthalmol Scand* 1996; 74: 558-62.
26. Seddon JM, Gragoudas ES, Egan KM, et al. Relative survival rates after alternative therapies for uveal melanoma. *Ophthalmology* 1990; 97: 769-77.
27. Zehetmayer M, Kitz K, Menapace R, et al. Local tumor control and morbidity after one to three fractions of stereotactic external beam irradiation for uveal melanoma. *Radiother Oncol* 2000; 55: 135-44.
28. Zimmerman LE, McLean IW, Foster WD. Does enucleation of the eye containing a malignant melanoma prevent or accelerate the dissemination of tumour cells. *Br J Ophthalmol* 1978; 62: 420-5.
29. Ghazi NG, Ketcherside CS, Sheehan J, Conway BP. Gamma knife radiosurgery for uveal melanoma ineligible for brachytherapy by the Collaborative Ocular Melanoma Study criteria. *Open Access Surgery* 2008; 1: 21-4.
30. Krema H, Somani S, Sahgal A, et al. Stereotactic radiotherapy for treatment of juxtapapillary choroidal melanoma: 3-year follow-up. *Br J Ophthalmol* 2009; 93: 1172-6.
31. Gragoudas ES, Lane AM, Regan S, et al. A randomized controlled trial of varying radiation doses in the treatment of choroidal melanoma. *Arch Ophthalmol* 2000; 118: 773-8.
32. Furdova A, Strmen P, Sramka M. Complications in patients with uveal melanoma after stereotactic radiosurgery and brachytherapy. *Bratisl Lek Listy* 2005; 106: 401-6.
33. Mosci C, Mosci S, Barla A, Squarcia S, Chauvel P, Iborra N. Proton beam radiotherapy of uveal melanoma: Italian patients treated in Nice, France. *Eur J Ophthalmol* 2009; 19: 654-60.
34. Girkin CA, Comey CH, Lunsford LD, Goodman ML, Kline LB. Radiation optic neuropathy after stereotactic radiosurgery. *Ophthalmology* 1997; 104: 1634-43.
35. Isager P, Ehlers N, Urbak FS, Overgaard J. Visual outcome, local tumour control, and eye preservation after 106Ru/Rh brachytherapy for choroidal melanoma. *Acta Oncol* 2006; 45: 285-93.
36. Henderson MA, Shirazi H, Lo SS, et al. Stereotactic radiosurgery and fractionated stereotactic radiotherapy in the treatment of uveal melanoma. *Technol Cancer Res Treat* 2006; 5: 411-9.